



TITLE PAGE

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- ❖ Theme : AI (DIVYA DRISHTI)
- ❖ Problem Statement Title : Sharodshav
- ❖ PS Category : Software



SHARODSHAV

Smart Digital Platform for Kolkata's Durga Puja Experience



Proposed Solution :

- **SHARODSHAV** is a smart digital platform designed to make Kolkata's Durga Puja experience easier, more accessible and more engaging by combining **pandal discovery, interactive maps, trip planning, cultural information** and **AI-powered assistance** in **one platform**.



Key Features :



Pandal Discovery

Explore Durga Puja pandals through an interactive Kolkata map.



Trip Planner

Plan and organize Puja visits more conveniently.



Pujo AI

Get AI-based assistance for Puja-related queries and recommendations.



Nearby Places

Discover nearby attractions and useful locations.



Cultural Events

Access information about Puja events and celebrations.



Digital Participation

Experience features such as Virtual Pushpanjali and community interaction.



Problem Addressed :

- Puja information is **scattered** across different platforms.
- Visitors face difficulty discovering **pandals** and **planning routes**.
- Cultural events and **nearby facilities** are not available in **one place**.
- Users need a simple digital platform for both **planning** and **participation**.



Innovation and Uniqueness of the Solution :

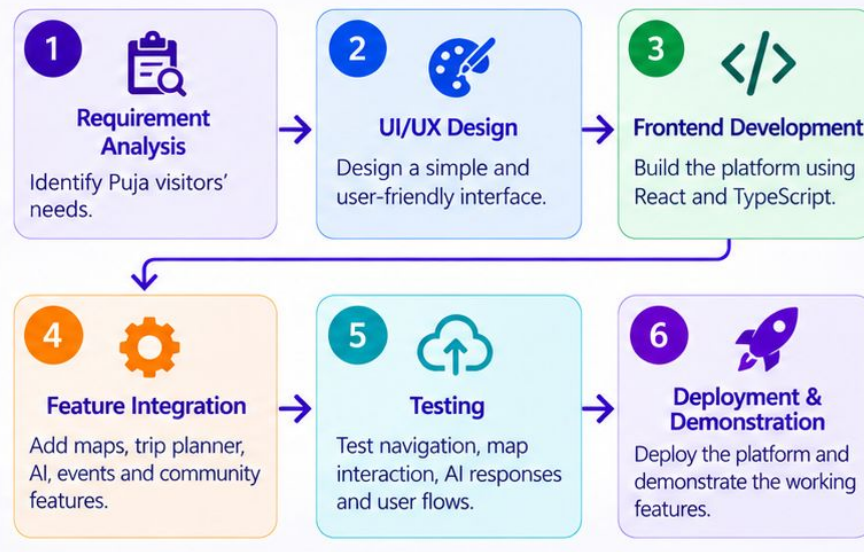
- Combines **technology, culture, AI and community interaction** in a single platform.
- Provides a **centralized and interactive** Sharodshav experience instead of relying on multiple sources.
- Offers **AI-powered assistance + real-time, location-based discovery** for a more personalized Puja experience.

TECHNICAL APPROACH

❖ Technologies Used

 Frontend	React + TypeScript + Vite	  
 Styling	Tailwind CSS	
 Maps	Leaflet + React-Leaflet	 
 Map Data	OpenStreetMap	
 Charts	Recharts	
 Icons	Lucide React	
 Data Handling	Structured Data + Local Storage	 
 Authentication/Services	Firebase	
 AI	Gemini API	
 Deployment	Web-based deployment	

❖ Implementation Methodology



❖ Technical Flow



❖ FEASIBILITY AND VIABILITY

A realistic, scalable and impactful solution for a smarter Sharodshav experience.



Analysis of the feasibility of the idea



Technical Feasibility

Uses modern, widely available technologies (React, Vite, Firebase, Maps API, Gemini API) – easy to develop and maintain.



Operational Feasibility

Can be managed by a small team with minimal infrastructure (cloud-based).



Economic Feasibility

Cost-effective using free tiers (Firebase, Google Maps, Gemini API).



Social Feasibility

High demand and real user value during Durga Puja; benefits visitors, organizers and local communities.



Feasible to build, deploy and scale with long-term impact.



Potential challenges and risks



Real-time & accurate data

Keeping pandal information, events and crowd data up to date.



High user traffic

During peak Puja days, the platform may face heavy load.



Location accuracy

Ensuring precise map locations and navigation for pandals.



AI reliability

Providing relevant and culturally appropriate responses.



Data privacy & security

Protecting user data and ensuring safe authentication.



Strategies for overcoming these challenges



Community-driven updates

Allow organizers and users to contribute and verify information.



Scalable cloud infrastructure

Use Firebase's scalable services to handle high traffic.



Optimized mapping

Use reliable APIs (Leaflet + OpenStreetMap + Esri) and continuous testing.



Prompt engineering & moderation

Fine-tune Gemini API with domain-specific prompts and filters.



Secure data practices

Use Firebase authentication, secure rules and follow data privacy guidelines.



With the right strategy, these challenges can be effectively managed.

❖ IMPACT AND BENEFITS

— Celebrating Culture. Connecting People. Creating Real Impact. —

❖ Potential impact on the target audience



Puja Visitors

Easily discover pandals, plan routes, get real-time updates and make the most of the Puja experience.

More
Joyful &
Hassle-free
Experience



Pandal Organizers

Better outreach, higher engagement and easy management of events, announcements and crowd information.

Greater
Visibility &
Participation



Local Businesses

Increased footfall and brand awareness during the festive season through digital listings and promotions.

Economic
Growth &
Opportunities



Local Communities

Stronger connection among people, preservation of cultural heritage and a more inclusive celebration.

A United &
Engaged
Community

❖ Benefits of the solution (social, economic, environmental, etc.)



Social Benefits

- Brings community together through a shared digital platform
- Promotes cultural heritage and tradition
- Improves accessibility for everyone (including elderly and disabled visitors)
- Encourages safe and well-informed celebrations



Economic Benefits

- Supports local businesses through digital visibility
- Increases footfall to pandals and events
- Cost-effective solution using free/cloud-based tools
- Creates opportunities for local artists, vendors and organizers.



Environmental Benefits

- Reduces use of paper by providing digital information
- Encourages use of public transport through location planning
- Helps manage crowd flow, reducing unnecessary travel and congestion
- Promotes a cleaner and more sustainable celebration.



Long-term Impact

- Scalable to other cities and festivals
- Can become a year-round cultural and event platform
- Strengthens digital infrastructure for local communities
- Builds a smarter, more connected and culturally aware society

❖ RESEARCH AND REFERENCES

Insights from trusted sources to build a meaningful and impactful solution.



Details / References

Official sources and tools used for our project



UNESCO – Durga Puja in Kolkata

Official recognition of Durga Puja as Intangible Cultural Heritage of Humanity (2021).

<https://ich.unesco.org/en/RL/durga-puja-in-kolkata-00703>Cultural
Heritage
Reference

UNESCO – Multimedia Archive

Cultural background, traditions, artisans and community participation.

<https://www.unesco.org/archives/multimedia/document-5613>Archive
Document

OpenStreetMap

Open geographic data used for map-based location and navigation features.

<https://www.openstreetmap.org>Geographic
Data Source

Leaflet

Open-source JavaScript library for interactive maps.

<https://leafletjs.com>Mapping
Library

Details / Research Works

Key studies and reports that support our solution



UNESCO ICH Research

Research on cultural heritage, community participation, crowd management and sustainability of Kolkata Durga Puja.

<https://ich.unesco.org/en/RL/durga-puja-in-kolkata-00703>Research
Study

UN India & UNESCO – Festival Accessibility

Guidelines for making Durga Puja more accessible for elderly people, persons with disabilities and all visitors.

<https://www.unesco.org/en/articles/un-india-and-unesco-launch-guidelines>Accessibility
Guidelines

Crowd & Cultural Accessibility (2026)

Research on navigation, crowd congestion, security coordination and improving cultural accessibility during Kolkata's Durga Puja.

<https://doi.org/10.29121/shodhkosh.v7.i12s.2026.7226>Research
Paper

Government of India – Cultural Documentation

Documentation on the heritage, cultural practices and inclusive nature of Durga Puja.

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=1781868>Government
Resource